



- 8 Fig. 8.1 shows part of the emission spectrum of visible radiation emitted by hydrogen gas in a star in a distant galaxy.

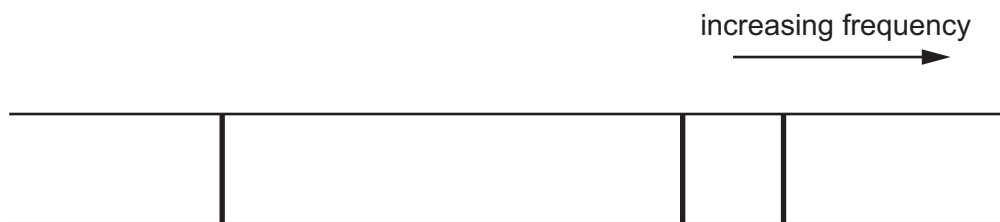


Fig. 8.1

The galaxy is moving away from the Earth at a speed of $6.2 \times 10^6 \text{ m s}^{-1}$.

- (a) (i) Explain how the positions of the lines in the emission spectrum seen by an observer on the Earth differ from the positions shown in Fig. 8.1.

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 [2]

- (ii) On Fig. 8.1, draw the three lines in possible positions in the spectrum seen by the observer. [2]

- (b) The lines in Fig. 8.1 correspond to electron transitions down to the energy level -3.40 eV . One of the lines represents emitted radiation of wavelength 488 nm .

- (i) Calculate the energy of a photon of this radiation.

photon energy = J [2]

- (ii) Determine the energy, in eV, of the energy level from which the electron transition originates to cause the emission of this radiation.

energy level = eV [2]





- (iii) Determine the wavelength, in nm, of this radiation as detected by the observer on the Earth.

wavelength = nm [2]

- (c) A value for the Hubble constant is $2.3 \times 10^{-18} \text{ s}^{-1}$.

Determine the distance of the galaxy from the Earth.

distance = m [2]

[Total: 12]

